

NUERBURG BARWEILER, Germany

WMO: 105060

Lat: 50.3603N

Lon: 6.8697E

Elev: 488

StdP: 95.60

Time Zone: 1.00 (EUC)

Period: 95-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.7	-7.7	-13.7	1.2	-7.1	-11.0	1.6	-5.3	11.5	6.0	10.2	5.8	2.9	100	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	27.6	17.9	25.5	17.2	23.7	16.6	19.1	25.1	18.2	23.5	17.4	22.1	3.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	13.0	20.9	16.3	12.3	20.2	15.5	11.7	19.4	56.3	25.2	53.4	23.6	50.7	22.0	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.3	8.0	7.1	DB	-12.7	31.0	3.3	2.1	-15.1	32.5	-17.0	33.8	-18.8	35.0	-21.2	36.5	
			WB	-13.2	20.5	3.2	0.7	-15.5	21.1	-17.4	21.5	-19.2	21.9	-21.6	22.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	0.4	1.2	4.2	7.9	11.9	14.7	16.7	16.6	13.1	9.2	4.5	0.8
	DBStd	7.04	4.39	4.41	4.21	4.14	3.87	3.65	3.41	3.33	3.26	3.79	3.63	4.11
	HDD10.0	1366	298	247	183	88	26	3	0	0	7	61	168	284
	HDD18.3	3664	557	481	438	312	203	120	76	76	162	284	416	542
	CDD10.0	803	0	0	4	26	83	143	206	204	99	36	2	0
	CDD18.3	59	0	0	0	0	2	10	23	21	3	0	0	0
	CDH23.3	479	0	0	0	1	18	81	175	184	21	1	0	0
	CDH26.7	97	0	0	0	0	1	11	38	47	1	0	0	0
Wind	WSAvg	3.4	3.6	3.8	3.6	3.6	3.2	3.0	3.2	2.8	3.0	3.4	3.4	3.6
Precipitation	PrecAvg	766	66	59	55	49	64	70	71	73	62	60	64	73
	PrecMax	931	182	144	111	104	120	146	167	163	143	148	103	193
	PrecMin	567	13	7	3	3	21	24	36	23	14	20	2	10
	PrecStd	93	34	31	30	23	26	29	32	40	35	28	23	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.5	11.9	17.1	22.3	25.6	28.5	29.9	31.2	25.8	20.4	13.8	10.5
		MCWB	9.3	8.1	10.0	12.4	16.1	18.7	18.4	18.9	16.5	15.2	10.9	9.1
	2%	DB	9.2	9.7	14.2	19.8	22.5	25.7	27.5	27.4	23.0	17.7	11.8	9.2
		MCWB	8.1	6.9	8.8	12.0	15.1	17.7	17.9	17.9	16.4	13.9	10.1	8.2
	5%	DB	7.8	8.3	12.2	17.2	20.7	23.5	25.1	25.2	20.6	15.8	10.6	7.9
		MCWB	6.8	6.5	8.3	10.8	13.8	16.7	17.3	16.9	15.4	13.3	9.4	7.1
	10%	DB	6.2	7.1	10.2	14.8	18.7	21.4	23.2	23.0	18.5	14.2	9.3	6.5
		MCWB	5.2	5.7	7.5	9.6	13.1	15.8	16.6	16.4	14.5	12.3	8.3	5.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	9.1	11.2	14.0	18.0	19.9	20.1	20.1	18.1	15.9	11.7	9.7
		MCDB	10.1	10.6	14.4	19.6	23.1	25.6	26.3	27.3	22.3	18.8	13.2	10.2
	2%	WB	8.1	7.9	9.9	12.6	16.2	18.5	19.1	18.9	16.9	14.6	10.4	8.3
		MCDB	8.9	9.0	12.9	18.0	20.9	23.7	24.9	25.3	21.4	16.7	11.5	9.0
	5%	WB	6.9	6.9	8.9	11.5	14.8	17.5	18.2	17.9	15.9	13.5	9.5	7.1
		MCDB	7.7	8.0	11.6	16.0	19.4	22.3	23.6	22.8	19.8	15.4	10.6	7.8
	10%	WB	5.3	5.8	7.8	10.3	13.6	16.4	17.2	17.0	14.8	12.5	8.4	5.7
		MCDB	6.1	6.9	9.7	14.3	17.5	20.5	21.9	21.6	18.0	14.1	9.3	6.4
Mean Daily Temperature Range	5% DB	MDBR	4.1	4.8	6.4	7.9	8.2	8.5	8.4	8.6	7.4	6.0	4.3	3.9
		MCDBR	5.0	6.6	9.7	11.4	11.5	11.8	11.7	12.5	10.7	7.8	5.5	4.3
	5% WB	MCWBR	4.5	4.6	5.6	5.6	5.9	5.7	4.9	5.2	5.4	4.6	4.2	4.0
		MCDDBR	4.8	5.2	7.7	10.0	10.4	10.5	10.4	10.4	9.2	7.0	4.9	4.1
		MCWBR	4.7	4.2	5.0	5.2	5.7	5.6	4.9	5.1	5.0	4.6	4.0	4.1
Clear-Sky Solar Irradiance	taub	0.300	0.324	0.356	0.388	0.394	0.396	0.397	0.392	0.363	0.345	0.320	0.297	
	taud	2.443	2.392	2.324	2.272	2.284	2.293	2.284	2.328	2.389	2.437	2.441	2.428	
	Ebn at Noon	744	812	843	853	863	861	854	840	827	776	707	687	
	Edh at Noon	60	81	103	122	126	126	125	114	96	77	60	53	
All-Sky Solar Radiation	RadAvg	0.77	1.47	2.60	4.14	4.86	5.37	5.08	4.37	3.25	1.89	0.87	0.59	
	RadStd	0.12	0.28	0.40	0.49	0.47	0.51	0.59	0.43	0.41	0.19	0.11	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	+0.71	+0.78	N/A	N/A	N/A
(47) Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page